



Linux OS – Lab 5

Task 1: Linux File Permissions

1. Create a file named **script.py** and give everyone (u, g, o) **write** only permission.
2. Try to display the contents (e.g., **cat**) of the **script.py** file, what is the output, explain?
3. Remove **write** permissions for **others**.
4. Make **script.py** executable for the **owner**.
5. Create a directory **Project**, then give **read**, **write**, and **execute** permissions to **all** users.
6. Reset **Project** so that only the **owner** has **read** and **write** access.
7. Create a **new.txt** file with **444** permissions, then attempt to:
 - Edit the file.
 - Remove the file.
8. Explain the difference between the "x" permission for **files** and **directories**.

Task 2: Linux File Ownership

1. Create a file named **report.txt**.
2. Change the owner of **report.txt** to **ahmed**.
3. Change the group ownership of **report.txt** to **admins**.
4. Verify the changes using **ls -l** command.
5. Change ownership of **report.txt** recursively for a directory **Reports**.



Task 3: Text Editors Usage

1. Create a file named **notes.txt** using **nano**.
2. Write some text (at least 6+ lines), save, and exit.
3. Display the contents of **notes.txt** using **cat**.
4. Open **notes.txt** in **vim**.
5. Copy the first line of the file then paste it at the bottom of the file.
6. Undo this change.
7. Delete three lines starting from the second line.
8. Search for the word "experience"
9. Save and exit vim.

Task 4: Setting a Default Editor

1. Check the current default editor using **\$EDITOR** environment variable.
2. Use **~/.bashrc** file to set **vim** as the default editor.
3. Verify the change by echoing the **EDITOR** variable.